

LEARNING MODULE

7

Company Analysis: Forecasting

LEARNING OUTCOMES

<i>Mastery</i>	<i>The candidate should be able to:</i>
☐	explain principles and approaches to forecasting a company's financial results and position
☐	explain approaches to forecasting a company's revenues
☐	explain approaches to forecasting a company's operating expenses and working capital
☐	explain approaches to forecasting a company's capital investments and capital structure
☐	describe the use of scenario analysis in forecasting

INTRODUCTION

1

Forecasts of companies' financial statements are used by analysts in valuation and to make investment recommendations. Developing the forecasts or projections is an important aspect of an analyst's job and is the focus of this module. In the first lesson, what to forecast, approaches to forecasting, and selecting a forecast horizon are discussed. The next three lessons focus on particular forecasts: revenues, operating expenses and working capital, and capital investments and capital structure. The final lesson discusses the use of scenario analysis in considering multiple outcomes.

LEARNING MODULE OVERVIEW



- Four common types of forecast objects are drivers of financial statement lines, individual financial statement lines, summary measures, and ad hoc objects. An analyst's choice of forecast object depends on available information, efficiency, accuracy, explanatory value, and verifiability.
- Forecast approaches generally are based on historical results, historical base rates and convergence, management guidance, or analyst discretion. An analyst's choice of forecast approach depends on the company's industry structure, sensitivity to the business cycle, and business model, as well as the reliability and availability of information.

- The choice of the forecast time horizon is determined by the investment strategy for which the security is being considered, the cyclicity of the industry, company-specific factors, and the analyst's employer's preferences.
- Revenue forecasts may be based on top-down or bottom-up forecast objects, using any of the four forecast approaches. Using different forecast objects and approaches to project revenue can be useful in uncovering implicit assumptions or errors in any single approach.
- Top-down revenue drivers include growth relative to GDP growth, and market growth and market share. Bottom-up revenue drivers include volumes and average selling prices; revenue by product line, geographic area, or reporting segment; capacity-based measures; and return-based measures.
- Analysts often use aggregated forecast objects or summary measures to forecast operating expenses because of a lack of disaggregated information. However, forecasts for operating expenses should be coherent with revenue forecasts. The choice of forecast object can vary depending on the forecast horizon.
- Working capital forecasts typically use efficiency ratios combined with revenue and operating expense forecasts to project accounts receivable, inventory, accounts payable, and other current assets and liabilities.
- Forecasts for capital expenditures may differentiate between maintenance and growth capital expenditures. Maintenance capital expenditure forecasts are often based on depreciation and amortization expenses. Growth capital expenditure forecasts are tied to a company's strategy, expansion plans, and revenue growth.
- Forecasts about a company's capital structure consider historical leverage ratios and capital structure, the company's financial strategy, and capital expenditure forecasts.
- Based on a company's risk factors, an analyst may develop several forecast scenarios rather than develop a single forecast. The analyst will judge the likelihood of each scenario occurring.

LEARNING MODULE SELF-ASSESSMENT



These initial questions are intended to help you gauge your current level of understanding of this learning module.

1. A benefit of using summary measures as forecast objects is *most likely*:
 - A. efficiency.
 - B. transparency.
 - C. explanatory value.

Solution:

A is correct. The benefit of using summary measures as a forecast object is efficiency, as fewer variables need to be forecast, but a disadvantage is less transparency. Forecasting financial statement lines or their drivers has the benefit of improved explanatory value.

2. Which of the following approaches is *most* appropriate for forecasting annual revenues for a company in a cyclical industry?

- A. Historical results
- B. Management guidance
- C. Analyst's discretionary forecast

Solution:

C is correct. An analyst's discretionary forecast is most appropriate, because it allows the analyst to consider both the current phase and the expected future phase of the business cycle. Historical results are a less appropriate forecast approach for companies in cyclical industries, because a future period is likely to be at a different point in the business cycle than the current or past period, and so results will differ. The use of guidance for companies that are highly sensitive to the business cycle is less appropriate, as management does *not* have an informational advantage over investors in forecasting macroeconomic variables like GDP or the prices of commodities.

3. An example of a top-down driver for revenue is a company's:

- A. market share.
- B. sales growth rate.
- C. products' average selling price.

Solution:

A is correct. Top-down drivers of revenue include the company's market share and the industry's market growth. The company's sales growth rate and products' average selling price are bottom-up drivers of revenue.

4. In developing a revenue forecast, non-recurring revenue *most likely*:

- A. has the same drivers as recurring revenue.
- B. will be disclosed by a company's management.
- C. is considered separately from recurring revenue.

Solution:

C is correct. Non-recurring items and effects should be considered separately. The non-recurring objects, such as changes in exchange rates or provisions for legal costs, generally do not have the same drivers as the recurring objects. While management may disclose some non-recurring items and effects, there are also non-recurring items and effects that management does not disclose, which require analyst judgment.

5. An analyst should consider the effects of changing commodity prices when forecasting:

- A. revenue only.
- B. cost of sales only.
- C. both revenue and cost of sales.

Solution:

C is correct. Both a company's cost of sales through input prices and its revenue may be affected by commodity prices. The effect on revenue depends on the company's competitive positioning and the price elasticity of demand for the company's products.

6. Working capital forecasts for a mature company with a unique business model are *most likely* made by:
- A. increasing the current assets and liabilities by the sales growth rate.
 - B. using the company's efficiency ratios combined with sales and operating forecasts.
 - C. using a historical base rate and convergence approach to develop forecasts of revenue, operating expenses, and efficiency ratios.

Solution:

B is correct. For a company with a unique business model, using a historical results approach is appropriate. Working capital forecasts are typically made by using efficiency ratios, which are combined with sales and operating cost forecasts to project accounts receivable, inventories, accounts payable, and other current assets and liabilities. The historical base rate and convergence approach is appropriate if the company is converging toward the norm for the industry. Adjusting current assets and liabilities by the sales growth rate is not appropriate, because changes in sales will affect each item differently.

7. Depreciation and amortization expenses are often used as the basis for:
- A. growth capital expenditures only.
 - B. maintenance capital expenditures only.
 - C. both growth and maintenance capital expenditures.

Solution:

B is correct. Maintenance capital expenditure forecasts are often based on depreciation and amortization expenses. Growth capital expenditure forecasts are more discretionary and are tied to management's expansion plans and revenue growth.

8. Which of the following generic risk factors is *most likely* to affect an established company in the consumer staples sector?
- A. Inflation or deflation
 - B. Technological developments
 - C. Changes in the business cycle

Solution:

A is correct. While all of these choices are generic risk factors, inflation or deflation is the one most likely to affect an established company in the consumer staples sector. The consumer staples sector is not likely to experience significant technological developments and is generally not sensitive to changes in the business cycle.

2

FORECAST OBJECTS, PRINCIPLES, AND APPROACHES



explain principles and approaches to forecasting a company's financial results and position

Approaches to forecasting vary by analyst and setting. Analysts who publish company research for external distribution tend to maintain quarterly forecasts and invest a great deal of effort to achieve forecast accuracy of revenue and earnings per share estimates. An investor with a controlling position in a private company has access to far more information, which may be used to build highly detailed models with a multi-year or even multi-decade perspective. Additionally, there are forecasting norms that vary by business model of the target company, such that models for banks appear markedly different from those for oil and gas producers.

In this module, we take the perspective of a longer-term-oriented analyst forecasting relatively straightforward public issuers—Warehouse Club Inc. and Iliso Marketplace Ltd., introduced in the prior two modules—but the principles discussed are broadly applicable. The more technical details of financial statement modeling using spreadsheets are covered in lessons in Financial Statement Analysis.

What to Forecast?

Analysts may focus on different **forecast objects** related to issuers' financial statements. Below are four common forecast objects.

1. *Drivers of financial statement lines*, which were illustrated in the earlier module on company analysis. Recall that for Warehouse Club Inc., net sales each year was analyzed using two bottom-up drivers: the average number of stores open and the average net sales per store, both of which were also analyzed using year-over-year percentage changes over the last 10 years. Forecasts of net sales (at the top of Warehouse's income statement) can be made by forecasting these drivers individually and multiplying them. Gross margin and SG&A expenses as percentages of net sales were identified as key drivers of operating expenses, which can be forecast and then multiplied by the net sales forecast to obtain the forecast cost of sales and SG&A expenses. Forecasting drivers rather than financial statement lines outright has the benefit of improved explanatory value and may improve accuracy; a financial statement line item can have multiple drivers moving in different directions, which might be difficult to forecast on an aggregated basis.
2. *Individual financial statement lines*. Rather than drivers, analysts can also directly forecast individual financial statement lines. This approach is often used for lines without clear drivers, for less-material items, and for items that the analyst does not have a perspective on. In some cases, the analyst may look for an estimate by management or simply assume the quantity will remain the same in future periods as in the past. Examples include lines such as amortization expense on the income statement (a forecast for which is often provided by management in the annual report), "other non-current assets" on the balance sheet (when small), and various lines on statements of cash flows for which minimal disclosures are provided.
3. *Summary measures* such as free cash flow, earnings per share, and total assets. The benefit of using these as forecast objects is efficiency. The disadvantage is less transparency, making it difficult to audit the forecast. This approach is most appropriate if the summary measure is stable and predictable, or if issuer disclosures are severely limited.
4. *Ad hoc objects*, which may not yet be reported on historical financial statements. In some cases, an analyst may need to forecast a loss or gain and its timing to make an investment decision with respect to the company's equity or debt securities before the issuer recognizes an accrual on its

financial statements. Examples of such cases include an issuer's announcing a material legal proceeding, government regulatory action, a tax dispute, or a natural disaster.

Focus on Objects That Are Regularly Disclosed

We recommend using forecast objects that either are disclosed regularly or can be directly calculated using what is disclosed regularly (e.g., for Warehouse Club: net sales, number of stores, net sales per store, national retail sales published monthly by a government agency). Information that is *not* disclosed regularly (such as the size of a market from a third-party consultancy's report) is suitable for *informing* forecasts but can be problematic for direct use because forecasts cannot be confirmed in a timely manner. While it may be intuitive to forecast sales and gross margin by individual product line for a company, if gross margin is disclosed only on a consolidated basis, it will be difficult for the analyst to verify product-line gross margin estimates. If the company's reported gross margin results differ from gross margin estimates, is the forecast off on product A, B, or C? Similarly, if a quantity is disclosed only on an annual basis, a forecast of it can be confirmed only once a year.

We also caution against overly complex models, because they require more forecasts and take more time to update, often without any improvement in accuracy. For example, the Swiss pharmaceutical company Novartis AG discloses revenues on a quarterly basis for over 35 individual drugs by US and Rest of World regions. Most of the drugs are approved for more than one type of illness. The company also has over 50 drugs in late-stage clinical development that may drive future revenue. While an analyst could forecast Novartis AG's revenues by forecasting sales for each of its individual drugs by illness and geography, it would take months to create such a model and weeks to update it every quarter, and the forecasts by illness are not verifiable over time based on the company's disclosures. Instead, the analyst should use judgment to focus on the most important drivers and use aggregations and shortcuts where the value-added of discrete forecasting is low. For example, it would not be a good use of an analyst's time to forecast sales for drugs with small addressable markets or those that have reached maturity and will exhibit slow growth until they lose patent protection.

Forecast Approaches

For any object, there are four general forecast approaches, as shown in Exhibit 1. In practice, they are often combined.

Exhibit 1: Forecast Approaches for Financial Statement Objects***Historical Results: Assume Past is Precedent***

This approach uses past observed or calculated values (such as a historical median) as a forecast. This is the easiest approach and is, perhaps, the default. Because the past occurred at least once before, it seems plausible to expect that it will again. However, past results were produced in past conditions, which may differ from current and future conditions.

USING HISTORICAL RESULTS AS FORECASTS

An analyst is forecasting the market size for energy drinks, a type of non-alcoholic ready-to-drink beverage. The analyst uses two drivers: a forecast of the market for non-alcoholic ready-to-drink beverages and energy drinks' share of that market. Sales of non-alcoholic ready-to-drink beverages have grown, on average, by 3% per year for the last 10 years. Since the analyst does not expect any material changes in conditions, she forecasts that the market will continue to grow by 3% per year.

The analyst believes that energy drinks bear some similarities to sports drinks, another type of non-alcoholic ready-to-drink beverage, which has achieved a market share of 20%. The analyst thus forecasts that energy drinks will also reach a market of 20% by 2035. These two forecasts are multiplied to arrive at a forecast market size for energy drinks.

This approach may be appropriate for companies operating in industries where the analyst does not expect the industry structure (e.g., Porter's Five Forces, PESTLE influences) to change, as well as for companies that have a low sensitivity to changes in the business cycle. This approach is also commonly used for forecast objects that are not material or that the analyst does not hold an opinion on.

Historical results are a less appropriate forecast approach for companies in cyclical industries, because a future period is likely to be at a different point in the business cycle than the current or past period, and so results will differ. An "over the cycle" average or median may be appropriate for a multi-year forecast for a cyclical company,

but less appropriate for a specific year as such an average or median disguises the year-to-year volatility. This forecast approach is also less appropriate for companies that are changing their competitive strategy or restructuring in some way, such as making a large acquisition or divestiture that renders historical results non-comparable.

Historical Base Rates and Convergence

This approach uses an industry or peer group average or median, computed over a long period of time, as a “base rate” for forecasting that an object will converge to over some time frame. The base rate could also be a macroeconomic variable such as GDP growth. This approach is not a rote historical exercise, as analyst discretion is required in selecting the object, the sample to calculate the base rate, and a time frame for convergence to the base rate.

HISTORICAL BASE RATES AND CONVERGENCE

Tesla is a relatively new entrant in the automotive industry, with an innovative suite of electric vehicles. An analyst may forecast Tesla’s income statement by forecasting that its sales growth, gross and operating margins, capital intensity, and capital structure will converge to historical automotive industry averages (using a peer group of publicly traded global automakers) over the next 10 years.

This forecast may be contested by other analysts who believe that the shift from internal combustion to electric vehicles, Tesla’s innovative battery technology, and its direct-to-consumer sales approach make historical auto industry averages irrelevant for Tesla.

Historical base rates and convergence may be appropriate for forecasting companies in well-established industries with many publicly traded peers, such as banks, airlines, restaurants, automakers, and retailers, especially when the analyst’s industry analysis does not call for material changes in industry structure or external industry influences. This approach is also logical for smaller companies that are “maturing into” a financial profile similar to that of larger peers with scale. For example, a biotechnology company that recently launched its first drug likely spends more than half its revenues on R&D, a ratio that is likely to decline toward the industry norm as the company scales.

This approach is less appropriate for companies in changing or new industries for which calculating a base rate is difficult or less relevant, for companies in highly cyclical industries for which a longer-term base rate and smooth convergence to it would obscure year-to-year volatility, and for companies that are industry leaders by a wide margin and account for a substantial proportion of any industry base rate. For example, “digital advertising industry” averages for sales growth and operating margins would be dominated by Alphabet and Meta, such that using them to forecast Alphabet’s and Meta’s financial results would simply be using their own historical results as the forecast, which may ignore any benefit from economies of scale (as the companies were smaller in the past).

Management guidance

Management of public companies may publicly provide targets for earnings, revenues, and other measures (e.g., capital expenditures) for the next quarter, year, or longer term, known as **management guidance** (or simply as guidance). Guidance can be detailed or rather directional and is often updated throughout the year. Initial guidance for next fiscal year might be provided during the fourth-quarter earnings call and updated for completed quarters, and new information provided at the first-, second-, and third-quarter earnings calls. Beyer et al. (2010) found that guidance accounts for a majority of the quarterly financial reporting information used by investors. Its

importance stems from its forward-looking rather than backward-looking nature, and because management has an abundance of industry and company information that investors do not.

Guidance is often provided as a range (e.g., “sales growth of 2%–4%”) and embeds many sub-forecasts and assumptions by management, including macroeconomic growth, cost inflation, market share changes, pricing actions, and changes in exchange rates. If management’s estimates for these change, so will guidance, and it is not uncommon for companies to suspend guidance altogether in periods of high uncertainty, such as during the COVID-19 pandemic or in recessions. A key focus of investors is understanding management’s assumptions embedded in guidance and scrutinizing their plausibility.

While the midpoint of a guidance range might be intuitively interpreted as management’s “true” expectations, Ciconte et al. (2013) found that the upper bound often better represents management’s expectations, which are “padded” by associating it with a pessimistic lower bound to create a hurdle that is more easily cleared, for which management can be rewarded.

SHORT- AND LONG-TERM MANAGEMENT GUIDANCE

Novo Nordisk A/S is a large biopharmaceutical company with shares and American Depositary Receipts (ADRs) listed on the NASDAQ Copenhagen and NYSE, respectively. Its management provides annual guidance that it updates each quarter, as well as less detailed but longer-term “strategic aspirations” during its annual Capital Markets Day presentations. Guidance includes management’s point and range estimates for the following (for each fiscal year):

- Sales growth
- Operating profit growth
- Net interest income/expense
- Effective tax rate
- Capital expenditures
- Depreciation and amortization
- Free cash flow

Using guidance for forecasts is appropriate when it is provided and when management has demonstrated a track record of reliable estimates (analysts should analyze past guidance versus actuals). We caution against the use of guidance for companies that are highly sensitive to the business cycle, as management does *not* have an informational advantage over investors at forecasting macroeconomic variables like GDP or the prices of commodities. Hutton et al. (2012) found that investors have an informational advantage at the macroeconomic level, while management has an advantage at the firm level and tends to make more accurate forecasts for objects under greater management control, such as operating expenses and capital expenditures.

Analyst’s discretionary forecast

We group all other forecast approaches as analysts’ discretionary forecasts, which include those based on surveys, quantitative models, probability distributions, analogies to historical precedents that differ from comparable companies or industry averages, and other unobservable inputs. This approach is most common for companies in cyclical industries, companies that have no or few comparables, those that do not provide management guidance, and/or those undergoing a fundamental change like a shift in the competitive or regulatory environment. For example, while shifts in energy production have happened before (e.g., coal to natural gas), climate change

and the potential shift to renewable energy and adoption of technologies like electric vehicles are unprecedented, with no historical observations to use as forecasts. Instead, an energy sector analyst must make discretionary forecasts based on many sources, such as government emission reduction commitments, proposed legislation, capital expenditure constraints, and so on.

DISCRETIONARY FORECASTS

An analyst is building a financial statement model for a medical technology company developing a new device to treat a type of stroke for which no other effective treatments exist. The analyst uses information from the device's clinical trials and epidemiology data from various medical journals to estimate the total addressable market of patients that are treatable with the device. The analyst also conducts a survey of specialist physicians to gauge their interest in using the device and to estimate the number of patients they would use it for (depending on the results of the clinical trials), and uses recent launches of other medical devices for neuro- and cardiovascular diseases to estimate market share for this device for the next 10 years. The analyst develops several market share scenarios based on the results of the clinical trials (i.e., the safer and more effective the device, the higher sales growth will be). While the analyst uses comparable, historical results as inputs, significant judgment is also required because this situation is unique.

Selecting a Forecast Horizon

The choice of the forecast time horizon is determined by the investment strategy for which the security is being considered, the cyclicity of the industry, company-specific factors, and the analyst's employer's preferences. Most professionally managed investment strategies describe the investment time frame and average holding period in the investment objectives of the strategy. For example, long-term-oriented fund managers may focus their forecasting primarily on a time period of three to five years, while shorter-term-oriented managers may focus more on the next one or two quarters.

The cyclicity of the industry could also influence the analyst's choice of time frame, because the forecast period should be long enough to allow the business to reach an expected mid-cycle level of sales and profitability. Similar to cyclicity, various company-specific factors, including recent acquisition or restructuring activity, can influence the selection of the forecast period to allow enough time for the realization of the expected benefits from such activity to be reflected in the financial statements. In other cases, there might be no individual analyst choice in the sense that the analyst's employer has specified fixed parameters.

QUESTION SET



1. A company discloses financial results by geographic area, and the economic exposure of each geographic area is similar. Explain why an analyst should or should not develop forecasts by geographic area and aggregate the results.

Solution:

An analyst should *not* develop forecasts by geographic area, because each area has similar economic exposure. Developing forecasts for each geographic area would require more time to develop and update, without a significant improvement in accuracy over an aggregate forecast.

2. Which approach is *most* appropriate for an analyst to use to forecast revenue for an industry leader in a mature, non-cyclical industry?

- A. Historical results
- B. Analyst's discretionary forecast
- C. Historical base rates and convergence

Solution:

A is correct. Historical results is appropriate for companies operating in mature industries where the analyst does not expect the industry structure to change and for companies with low sensitivity to the business cycle. Analyst's discretionary forecast is most common for companies that have no or few comparables, do not provide management guidance, and/or are undergoing a fundamental change. Historical base rates and convergence is not appropriate, because the company is an industry leader and would account for a substantial proportion of any industry base rate.

3. Which of the following statements about management guidance is *most* accurate?

- A. The availability of guidance increases during periods of high uncertainty.
- B. The midpoint of guidance best represents management's "true" expectations.
- C. Understanding the assumptions behind the guidance is critical to assessing the guidance.

Solution:

C is correct. A key focus of an analyst is to understand management's assumptions embedded in guidance and to scrutinize their plausibility. It is not uncommon for companies to suspend guidance altogether in periods of high uncertainty. While the midpoint of a guidance range might be intuitively interpreted as management's "true" expectations, management may provide a range specifically intended to create a hurdle that is more easily cleared, for which management can be rewarded.

4. Which of the following is an example of an ad hoc forecast object?

- A. Amortization expense
- B. Average sales per store
- C. Pending legal proceedings

Solution:

C is correct. Ad hoc objects are those that may not yet be reflected in historical financial statements. Amortization expense is an individual financial statement line. Average sales per store is a bottom-up driver of a financial statement line (revenue).

FORECASTING REVENUES

3



explain approaches to forecasting a company's revenues

After the forecast time horizon is determined, a revenue forecast requires the selection of the forecast objects and approach.

Forecast Objects for Revenues

Forecast objects for revenues are typically either top-down or bottom-up drivers, as discussed in the earlier module on company analysis. Common top-down forecast objects include “growth relative to GDP growth” and “market growth and market share.”

Growth relative to GDP growth. The analyst first forecasts the growth rate of nominal GDP. The analyst then considers how the growth rate of the specific company being examined will compare with nominal GDP growth. The analyst can use a forecast for real GDP growth to project volumes and a forecast for inflation to project prices. Analysts often think in terms of percentage point premiums or discounts derived from a company’s position in its life cycle (e.g., growth, mature, or decline) or business cycle sensitivity. Thus, an analyst’s conclusion might be that a healthcare company’s revenue will grow at a rate of 200 bps above the nominal GDP growth rate. The forecast could also be in relative terms. Thus, if GDP is forecast to grow at 4% and the company’s revenue is forecast to grow at a 50% faster rate, the forecast percentage change in revenue would be $4\% \times (1 + 0.50) = 6.0\%$, or 200 bps higher than the GDP growth rate forecast in absolute terms.

Market growth and market share. The analyst first forecasts a growth rate for a company’s product market, and then considers the company’s current market share and how that share is likely to change over time. For example, if a company is expected to maintain an 8% market share of a given product market and the product market is forecast to grow from CNY144 billion to CNY154 billion in annual revenue, the forecast growth in company revenue goes from a level of $8\% \times \text{CNY144 billion} = \text{CNY11.5 billion}$ to a level of $8\% \times \text{CNY154 billion} = \text{CNY12.3 billion}$ (considering this product market alone). If the product market revenue has a predictable relationship with GDP, regression analysis might be used to estimate the relationship.

In contrast, an analyst could select bottom-up drivers as the forecast objects, which are then aggregated to arrive at a forecast of total revenue for the company. Examples of bottom-up drivers for revenue forecasts include the following:

- *Volumes and average selling prices.* Forecasts for volumes and prices of the company’s products are prepared individually and multiplied to arrive at a revenue forecast. This approach is commonly used for companies that disclose these quantities, such as media and internet companies (users and revenues per user), airlines, asset managers (i.e., AUM and management fee rates), and commodity producers. Many companies disclose price and volume contributions to overall revenue growth, which can be used as the forecast object as well.
- *Product-line or segment revenues.* Forecasts for individual products, product or business lines, geographic areas, or reporting segments are made and then aggregated into a total revenue forecast. This approach is commonly used if a company makes such disclosures and if the objects have different economic exposures.
- *Capacity-based measures.* Forecasts, for example, in retailing, based on the number of stores and sales per store, or same-store sales growth (for stores that have been open for at least 12 months) and sales related to new-store openings.
- *Return- or yield-based measures.* Forecasts based on account balances and revenue yields on them. For example, net interest income for a bank can be calculated as loans multiplied by the average interest rate minus the product of deposits and liabilities and their average interest rate.

Using elements of both top-down and bottom-up objects can be useful for uncovering implicit assumptions or errors that could arise from using a single approach (e.g., forecasting revenues using bottom-up drivers but checking against what this approach would imply for the company's market size and share).

Separating Recurring and Non-Recurring Revenue or Revenue Growth

An important principle is that non-recurring items and effects should be excluded from a forecast object and considered separately. The non-recurring objects generally do not have the same drivers as the recurring object and may inflate or deflate its size. There are two types of non-recurring items and effects: those that are disclosed by a company's management and those that are not, which require analyst judgment to identify.

For revenue, the first type includes the effect of changes in exchange rates, extra selling days and weeks in the period, acquisitions/divestitures, and other "one-time" revenues or gains. These are separated from a revenue forecast object so that "underlying" revenue or growth can be forecast apart from the non-recurring items (for which a discrete forecast may not be made at all). This is generally a straightforward task unless the credibility of an issuer's financial reporting is in doubt, in which case the analyst may judge certain "non-recurring" items as, in fact, recurring. Some analysts may also use proprietary exchange rate forecasts in revenue projections.

The second type of non-recurring items is those *not* quantified by management and that require analyst judgment to estimate. For example, the COVID-19 pandemic had numerous economic effects, including a rapid increase in e-commerce sales as a percentage of retail sales in most geographies. Many believed that this effect was a "new normal," that the increase in e-commerce sales would recur and subsequent periods would show growth from the newly achieved level. By the second half of 2021, this belief was called into question, because e-commerce sales as a percentage of retail sales began to decline, returning to a trend established prior to the pandemic. Many e-commerce companies saw revenue decelerate or decline in 2022, confirming that some amount of e-commerce's COVID-19-related growth was, in fact, non-recurring. As a second example, graphic processing units (GPUs) are types of semiconductors used in a variety of applications, including gaming PCs. GPU manufacturers saw a large increase in sales in the late 2010s, which many believed was partly attributable to the use of GPUs in cryptocurrency mining, but the companies either could not or would not quantify the precise amount. Although that amount is unknown, an analyst may judge such sales as non-recurring (e.g., due to government restrictions on cryptocurrency mining) and forecast these GPU sales separately from those attributable to gaming.

Forecast Approaches for Revenues

Any of the four forecast approaches introduced in Exhibit 1 in the prior lesson (historical results, historical base rates and convergence, management guidance, discretionary forecasts) can be used for forecasting revenue objects, with the selection based on the analyst's judgment, the information environment, and the issuer's business model. We illustrate how Elaine Nguyen might use the four approaches to forecast revenues for Warehouse Club Inc. in Exhibit 2.

Exhibit 2: Forecast Approaches for Warehouse Club Inc.'s Revenues

Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Top-down drivers (e.g., market size, market share)	Historical results	US retail sales excluding autos has grown on average by 3.4% each year for the last 10 years and the company's share of that (70 basis points in 2X19) has increased by 2 basis points per year. Nguyen can forecast revenues by assuming 3.4% market growth and 2 basis points of market share gain each year going forward.
	Historical base rates and convergence	Rather than assuming continued share gains, Nguyen may forecast that Warehouse's sales growth rate will converge to the market growth rate over the next five years, and forecast revenues by assuming a smooth decline in annual market share gains to 0 basis points by 2X25.
	Management guidance	If Warehouse provides annual or longer-term expectations for market size, market share, or sales, Nguyen can use these in her revenue forecast.
	Analyst's discretionary forecast	Warehouse has operated stores in some regional geographies for a long time. Nguyen may commission a study or survey to determine Warehouse's market share in its most established markets (which could be multiples of its all-in market share) and develop market size and share estimates by established/growing/new regions for Warehouse and aggregate them into a sales forecast.
Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Bottom-up drivers (e.g., stores, sales per store, membership, membership annual price)	Historical results	Warehouse has opened 4–6 new stores, grown net sales per store by 3.8%, and added ~450,000 members per year for the last 10 years. Nguyen could assume these same quantities going forward as the revenue forecast.
	Historical base rates and convergence	There are several mature retailers in the USA that ceased opening new stores several years ago. Nguyen may slow the rate of new-store openings gradually to zero in the USA.
	Management guidance	Management has provided guidance on new-store openings per year and the annual price of membership, which are bottom-up drivers for the revenue forecast.
	Analyst's discretionary forecast	Nguyen may develop a discretionary forecast for stores and members by determining stores per capita and the ratio of members to stores in Warehouse's more mature regional geographies and then applying these quantities, or fractions of them, to different regions where Warehouse is opening stores today. This approach would require a geographic analysis of stores and commissioning surveys to determine regional membership counts or the use of third-party or government data, if available.

Analysts must incorporate their view (implicitly or explicitly) on key risk factors in their revenue forecasts. Four risk factors to consider for all companies are competition, changes in the business cycle, inflation and deflation, and technological developments. For Warehouse Club Inc., two key risk factors are competition and technological developments, embodied in e-commerce, which has been taking share in US retail for many years. Views on these risk factors are also embedded in forecasts of profitability,

working capital, capital investments, and capital structure. Given the potential wide range of outcomes, analysts often construct several forecasts for an issuer's financial statements called *scenarios*, each based on different views of key risk factors. We discuss scenario analysis in the last lesson of this module, after forecasting other key financial statement drivers.

QUESTION SET



An analyst gathers the following information for Iliso Marketplace Ltd. (Iliso), the e-commerce retailer introduced in earlier modules. Recall that sales by third-party merchants are recognized on a net basis, because Iliso does not take control of the inventory. Iliso receives 15% of the gross merchandise value (GMV) of third-party sales as commission.

	2X19 (USD millions)
National retail sales excluding autos	4,165,000
Iliso's retail sales	7,336
Iliso's third-party merchant sales	1,269
Total GMV	8,605
Iliso's retail sales	7,336
Third-party merchant fees	190
Total Revenues	7,526

Gross merchandise value has increased by an average of 3.6% per year from 2X15 to 2X19 and by 2.7% from 2X18 to 2X19. Iliso's GMV as a percentage of national retail sales has increased steadily, from 0.08% in 2X15 to 0.21% in 2X19.

	2X19	Average Growth per Year, 2X15 to 2X19	Growth from 2X18 to 2X19
Average sales/customer account (USD)	204	5.9%	4.6%
# accounts (millions)	36	22.5%	16.1%
Iliso's retail sales (USD millions)	7,336	29.7%	21.4%
Average sales/third-party merchant (USD)	31,300	13.6%	9.8%
# third-party merchants	40,500	28.5%	28.6%
Third-party merchant sales (USD millions)	1,269	46.0%	41.2%

1. Explain how an analyst would use top-down drivers to forecast Iliso's 2X20 GMV.

Solution:

An analyst would first forecast national retail sales excluding autos and then forecast Iliso's GMV as a percentage of that market, which represents Iliso's market share. National retail sales excluding autos have been increasing

each year, and the analyst is likely to use a growth rate between 2.7% and 3.6%, depending on economic conditions. Iliso's market share in 2X19 was 0.21%, and given past increases, an increase of 2 to 3 bps seems reasonable.

2. Explain why an analyst using top-down drivers needs to allocate Iliso's forecast GMV sales between its own retail sales and sales by third-party merchants. Describe two alternative approaches to doing so.

Solution:

It is important that an analyst allocate the forecast total GMV between Iliso and third-party merchants, because Iliso's revenue and cost structure differs for each. Iliso recognizes revenue on a gross basis for its own retail sales and on a net basis for sales made by third-party merchants. Iliso's take rate is 15% on third-party merchant sales, but its gross margin percentage is far greater.

Currently, Iliso's retail sales account for 85% of its GMV, while third-party merchant sales account for the remaining 15%. An analyst could allocate the forecast increase in GMV sales based on those percentages. Alternatively, an analyst could allocate an increasing proportion to third-party merchant sales, because they have been growing at a faster rate than Iliso's retail sales.

3. Given the following assumptions, the analyst's forecast of Iliso's 2X20 total revenues is *closest* to:

- National retail sales excluding autos grows by 3.4%.
- Iliso's market share (i.e., its GMV as a percentage of national retail sales excluding autos) increases by 2 bps.
- GMV is split 75%/25% between retail sales and third-party merchant sales.
- The take rate on third-party merchant sales remains the same as in 2X19.

A. 7,800.

B. 8,441.

C. 9,906.

Solution:

A is correct.

Forecast of national retail sales excluding autos = $1.034 \times 4,165,000$
= 4,306,610

Forecast of Iliso's 2X20 GMV = $4,307,000 \times (0.21\% + 2 \text{ bps}) = 9,906$

Forecast of Iliso's 2X20 retail sales = $9,905 \times 75\% = 7,430$

Forecast of 2X20 third-party merchant sales = $9,905 \times 25\% = 2,477$

Forecast of 2X20 third-party merchant fees = $2,476 \times 15\% = 371$

Forecast of Iliso's 2X20 total revenues = $7,429 + 371 = 7,801$

4. An analyst who uses bottom-up drivers forecasts the number of customer accounts to increase by 15%, the average sales per customer to increase by 5%, the number of third-party merchants to increase by 28%, and the aver-

age sales per third-party merchant to increase by 40%. The analyst's forecast for Iliso's 2X20 GMV (in USD millions) is *closest* to:

- A. 8,860.
- B. 9,210.
- C. 11,140.

Solution:

C is correct.

Forecast of Iliso's 2X20 retail sales = $204(1.05) \times 36(1.15) = 8,868$ or = $7,336(1.05)(1.15)$
= 8,858

Forecast of 2X20 third-party merchant sales = $[40,500(1.28) \times 31,300(1.40)]/1,000,000 = 2,272$ or = $1,269(1.28)(1.40)$
= 2,274

Forecast of Iliso's 2X20 GMV = $8,868 + 2,272 = 11,140$

FORECASTING OPERATING EXPENSES AND WORKING CAPITAL

4

- ☐ explain approaches to forecasting a company's operating expenses and working capital

Issuers' disclosures about operating costs are typically less detailed than revenue disclosures. Rather than modeling costs separately for different geographic regions, business segments, or product lines, analysts are often forced to use more aggregated forecast objects such as consolidated financial statement lines (e.g., cost of sales, SG&A) or summary measures like EBITDA margins on a consolidated or segment basis. The forecasts for revenues and costs should still be coherent. If sales of a relatively low-margin product, segment, or geography are forecast to grow faster than other revenues, some level of overall profit margin deterioration should be forecast, even if the analyst is uncertain about the precise margins earned on each object. For example, food and grocery typically has low gross profit margins, so an analyst of a grocery chain that also sells higher-margin items, such as alcoholic products or pharmaceutical products, would want to forecast any change in product mix sold.

Cost of Sales and Gross Margins

Cost of sales (cost of goods sold, or COGS) is typically the single largest cost for companies that make and/or sell products. Because it has a direct link with sales, forecasting this item as a percentage of sales (or as a gross margin) is usually a good approach. If a company is losing market share to new substitute products with lower prices, gross margins are likely to decline. But if the company is gaining market share because it has introduced differentiated products—especially if it has done so in combination with achieving cost advantages—gross margins are likely to rise.

Cost of sales is typically a large cost, so even a single basis point in a gross margin forecast can have a material impact on forecasts of operating profit and free cash flow. Analysts should consider whether a more detailed analysis of these costs (e.g., by segment, input, or product line, or by volume and price components) is possible to better justify the forecast. For example, some companies face fluctuating input costs that can be passed on to customers only with a time lag. Particularly for companies with low gross margins, sudden shocks in input costs can affect operating profit significantly. A good example is the sensitivity of airlines' profits to unhedged changes in jet fuel costs. In these cases, a breakdown of both costs and sales into volume and price components is essential for developing short-term forecasts, even if analysts use the overall relationship between sales and input costs for developing longer-term forecasts, as shown in Exhibit 3.

Exhibit 3: The Effect of Prices and Costs on Gross Profit and Margin

Assume that a company's COGS as a percentage of sales equals 25% and that the quantity sold is the same in Period 2 as in Period 1. If input costs double in Period 2 and the company can pass the entire increase on to its customers through a 25% price increase, COGS as a percentage of sales will increase (to 40%) because an equal absolute amount has been added to the numerator and to the denominator.

	Period 1	Period 2
Sales	100.0	125.0
COGS	25.0	50.0
Gross profit	75.0	75.0
COGS as % of sales	25%	40%
Gross margin %	75%	60%

Thus, although the absolute amount of gross profit will remain constant, the gross margin will decrease (from 75% to 60%).

Analysts should incorporate a company's hedging strategy into a forecast. For example, commodity-driven companies' gross margins almost automatically decline if input prices of needed commodities increase significantly, because the variable costs increase at a faster rate than output prices. Through hedging strategies, a company can mitigate the impact on profitability. For example, brewers often hedge the cost of barley, a key raw material needed for brewing beer, one year in advance. Although companies usually do not disclose specific hedging positions, their hedging strategy is often disclosed in the notes to the financial statements. Further, the negative impact of increasing sales prices on sales volume (if product demand is price elastic) can be mitigated by a policy of gradual sales price increases. For example, if the brewer expects higher barley prices because of a poor harvest, the brewer can slowly increase prices to avoid a strong price jump next year.

While competitors' gross margins can provide a useful cross-check for forecasting gross margin, the margins may not be comparable given differences in business models. For example, some retailers own and operate their own stores, whereas other companies operate as wholesalers with franchised retail operations. In the franchise model, most of the operating costs are incurred by the franchisee; the wholesaler sells products with only a small markup to these franchisees. Compared with a retailer with its own stores, a wholesaler will have a much lower gross margin but also much lower operating costs.

SG&A Expenses

SG&A expenses are the other main type of operating costs. In contrast to cost of sales, SG&A expenses often have a less direct relationship with revenues. As an illustration of the profit impact of COGS and SG&A, consider the case of a Thai cement and building materials company, Siam Cement Group, from 2017 to 2018. A summary of the company's key income statement items is shown in Exhibit 4.

Exhibit 4: Siam Cement Group 2017–18 Financials

	2017 (THB billions)	2018 (THB billions)	YoY%	Percentage of Sales	
				2017	2018
Net sales	450.92	478.44	6.1	100.0	100.0
COGS	349.31	383.46	9.8	77.5	80.1
Gross profit	101.61	94.98	−6.5	22.5	19.9
SG&A	52.58	55.09	4.8	11.7	11.5
<i>Selected SG&A items:</i>					
Salary and personnel expenses	24.24	23.98	−1.1	5.4	5.0
Freight costs	11.63	11.55	−0.7	2.6	2.4
Research and development	4.18	4.67	11.7	0.9	1.0
Promotion and advertising	2.62	2.58	−1.5	0.6	0.5
Operating income	49.03	39.89	−18.6	10.8	8.4

Note: “YoY%” means year-over-year percentage change.

Source: Based on information in Siam Cement Group's annual reports.

As shown in Exhibit 4, Siam Cement was affected in 2018 by higher input costs that could not be fully passed on to customers. Consequently, despite sales growth of 6.1%, gross profit fell by 6.5% and gross margin declined. The company was able to limit its other operating costs; SG&A expenses grew 4.8%, declining slightly as a percentage of revenue. Operating income fell 18.6%. This experience contrasts with the company's experience in 2016 (not shown), when lower input costs resulted in a widening of the gross margin by 240 basis points versus the prior year.

Although SG&A expenses overall are generally less closely linked to revenue than is cost of sales, certain SG&A expenses could be more variable than others. Specifically, selling and distribution expenses often have a large variable component and can be modeled as a percentage of sales, while general corporate costs are more fixed and might be better modeled using a fixed growth rate based on expected wage inflation.

Segment disclosures often include operating and EBITDA margin profitability by segment, but generally do not include cost items such as cost of sales, SG&A, etc., by segment. If an analyst is constructing a model based on segment forecasts, the analyst might instead use summary measures on a segment basis, as shown in Example 1.

EXAMPLE 1**Analysis of a Consumer Goods Company with Multiple Segments**

A consumer goods company reported an overall underlying operating margin of 19.1% in 2X19. As shown in Exhibit 5, the operating margin is lowest in its fastest-growing segment, Home Care. The other two segments enjoy higher margins but are growing more slowly.

Exhibit 5: Segment Revenue and Profits (EUR millions, unless noted)

Segment	2X19	2X18	'19/'18 YoY	Avg. Growth Rate, 2X17–2X19
Beauty & Personal Care	21,868	20,624	6%	3%
Foods & Refreshments	19,287	20,227	–5%	–5%
Home Care	10,825	10,131	7%	3%
Total revenues	51,980	50,982	2%	0%
<i>Underlying operating profit</i>				
Beauty & Personal Care	4,960	4,543	9%	
Foods & Refreshments	3,382	3,576	–5%	
Home Care	1,605	1,344	19%	
Total	9,947	9,463	5%	
<i>Underlying operating profit margin</i>				
Beauty & Personal Care	22.7%	22.0%		
Foods & Refreshments	17.5%	17.7%		
Home Care	14.8%	13.3%		
Total	19.1%	18.6%		

Note: Underlying operating profit is a non-IFRS operating profit measure, equal to IFRS operating profit adjusted for items such as restructuring costs.

Source: Example adapted from Unilever.

1. Determine the estimated sales, operating profit, and operating profit margin by using the following two approaches: (A) Assume total sales growth of 2.0% and overall underlying operating margin of 19.1% for the next five years, and (B) assume each individual segment's sales growth and underlying operating margin continue at the same rate reported in 2019. Which approach will result in a higher estimated underlying operating profit after five years?

Solution:

Exhibit 6 shows that operating profit after five years will be EUR10,962 million under Approach A and EUR11,549 million under Approach B.

Exhibit 6: Sales and Operating Profit, 2X19–2X24E (EUR millions, unless noted)

Approach A	2X19A	2X20E	2X21E	2X22E	2X23E	2X24E
Total revenues	51,980	53,020	54,080	55,162	56,265	57,390
Growth rate		2%	2%	2%	2%	2%
Underlying operating profit	9,947	10,127	10,329	10,536	10,747	10,962
Growth rate		2%	2%	2%	2%	2%
Underlying operating profit margin	19.1%	19.1%	19.1%	19.1%	19.1%	19.1%
Approach B						
Sales	2X19A	2X20E	2X21E	2X22E	2X23E	2X24E
Beauty & Personal Care	21,868	23,187	24,586	26,069	27,641	29,308
Growth rate		6%	6%	6%	6%	6%
Foods & Refreshments	19,287	18,391	17,536	16,721	15,944	15,203
Growth rate		–5%	–5%	–5%	–5%	–5%
Home Care	10,825	11,567	12,359	13,205	14,110	15,077
Growth rate		7%	7%	7%	7%	7%
Total revenues	51,980	53,144	54,481	55,995	57,695	59,588
Margins						
Beauty & Personal Care	22.7%	22.7%	22.7%	22.7%	22.7%	22.7%
Foods & Refreshments	17.5%	17.5%	17.5%	17.5%	17.5%	17.5%
Home Care	14.8%	14.8%	14.8%	14.8%	14.8%	14.8%
Underlying operating profit						
Beauty & Personal Care	4,960	5,259	5,576	5,913	6,269	6,648
Foods & Refreshments	3,382	3,225	3,075	2,932	2,796	2,666
Home Care	1,605	1,715	1,832	1,958	2,092	2,235
Total	9,947	10,199	10,484	10,803	11,157	11,549

2. Compare and explain the results under the two alternative approaches (A and B) in Question 1.

Solution:

Approach A assumes a constant 2% total sales growth rate and a stable 19.1% underlying operating margin. Thus, the operating profit growth rate is in line with the revenue growth rate and constant at 2%, which therefore assumes no difference in growth rates and profitability of the segments. Approach B assumes growth rates of 6%, –5%, and 7% of sales for the Beauty & Personal Care, Foods & Refreshments, and Home Care segments. This approach results in a faster overall compounded growth rate than under Approach A (3% versus 2%) and an annual increase, on average, in the total underlying operating profit margin of 6 bps due to the mix effect of different segment margins. In 2024E, Approach A yields an underlying operating profit margin of 19.1%.

3. Assume the company can grow segment revenues over the next five years at the following rates: Beauty & Personal Care, 3.0%; Foods & Refreshments, 2.0%; Home Care, 4.0%. But underlying operating profit margins in Beauty & Personal Care will fall 20 bps annually for the next five years (because of high competition, limited growth, and costs resulting from the adoption of sustainable packaging), and operating profit margins in the Foods & Refreshments and Home Care segments will increase by 15 and 50 bps, respectively, each year for the next five years (helped by increasing demand for the company's products and better utilization of its factories). Calculate the overall underlying operating profit margin in each of the next five years.

Solution:

As shown in Exhibit 7, the overall underlying operating profit margin improves from 19.1% in 2X19 to 19.5% in 2X24, because the margin decline in Beauty & Personal Care is more than offset by the margin increase in Foods & Refreshments and the faster-growing Home Care segment.

Exhibit 7: Sales and Operating Profit, 2X19–2X24E (EUR millions, unless noted)

Sales	2X19A	2X20E	2X21E	2X22E	2X23E	2X24E
Beauty & Personal Care	21,868	22,524	23,200	23,896	24,613	25,351
Growth rate		3%	3%	3%	3%	3%
Foods & Refreshments	19,287	19,673	20,066	20,468	20,877	21,294
Growth rate		2%	2%	2%	2%	2%
Home Care	10,825	11,258	11,708	12,177	12,664	13,170
Growth rate		4%	4%	4%	4%	4%
Total revenues	51,980	53,455	54,974	56,540	58,153	59,816
Margins	2X19A	2X20E	2X21E	2X22E	2X23E	2X24E
Beauty & Personal Care	22.7%	22.5%	22.3%	22.1%	21.9%	21.7%
Foods & Refreshments	17.5%	17.7%	17.8%	18.0%	18.1%	18.3%
Home Care	14.8%	15.3%	15.8%	16.3%	16.8%	17.3%
Underlying operating profit	2X19A	2X20E	2X21E	2X22E	2X23E	2X24E
Beauty & Personal Care	4,960	5,064	5,169	5,277	5,386	5,496
Foods & Refreshments	3,382	3,479	3,579	3,681	3,786	3,894
Home Care	1,605	1,725	1,853	1,988	2,131	2,282
Total	9,947	10,268	10,601	10,946	11,303	11,672
Margin	19.1%	19.2%	19.3%	19.4%	19.4%	19.5%

Working Capital Forecasts

Working capital forecasts are typically made by using efficiency ratios, discussed in earlier modules, as the forecast object, which are combined with sales and costs forecasts to project accounts receivable, inventories, accounts payable, and other current assets and liabilities, as demonstrated in Example 2.

EXAMPLE 2**Working Capital Forecasts with Efficiency Ratios**

Exhibit 8 shows revenue, COGS, and year-end working capital account balances for YY Ltd., a fictional company, for years 1–3. Based on the data in the exhibit, answer Questions 1–3.

Exhibit 8: YY Ltd. Financial Data (CNY millions)

Year	1	2	3
Revenue	174,915	205,839	245,866
COGS	152,723	177,285	209,114
Accounts receivable	5,598	6,949	10,161
Inventory	29,481	32,585	41,671
Accounts payable	46,287	59,528	72,199

1. Calculate days sales outstanding (DSO), inventory days on hand (DOH), and days payable outstanding (DPO) for years 1, 2, and 3, using year-end balances and assuming a 365-day fiscal year.

Solution:

Recall that days sales outstanding is equal to accounts receivable/(revenues/365), inventory days on hand is equal to inventories/(COGS/365), and days payable outstanding is equal to accounts payable/(COGS/365). Using the data in Exhibit 8, the three ratios for years 1–3 are as follows:

Year	1	2	3
Days sales outstanding	12	12	15
Inventory days on hand	70	67	73
Days payable outstanding	111	123	126

2. Your colleague Liang forecasts revenue growth of 18%, 16%, and 13% and gross margins of 17%, 17%, and 16% in years 4, 5, and 6, respectively, for YY Ltd. Using Liang's forecasts, calculate expected revenue and COGS in years 4, 5, and 6.

Solution:

Using Liang's forecasts for annual revenue growth and gross margins and the data in Exhibit 8, expected revenue and COGS for years 4–6 are as follows:

Year (CNY millions)	1	2	3	4	5	6
Revenue	174,915	205,839	245,866	290,122	336,541	380,292
Growth rate		18%	19%	18%	16%	13%
COGS	152,723	177,285	209,114	240,801	279,329	319,445
Gross margin	13%	14%	15%	17%	17%	16%

3. Liang forecasts that days sales outstanding, inventory days on hand, and days payable outstanding in years 4, 5, and 6 will remain the same as the year 3 amounts. Using Liang's forecasts as well as forecast revenue and COGS, calculate expected accounts receivable, inventories, and accounts payable year-end balances for years 4, 5, and 6.

Solution:

Each of the efficiency ratios can be rearranged to yield working capital balances, because we have values for two of the three variables in them: the efficiency ratios are assumed to remain constant from year 3 levels, and revenue and COGS have already been forecast.

Days sales outstanding is equal to accounts receivable/(revenues/365); thus, accounts receivable is equal to days sales outstanding $\times$ (revenues/365).

Similarly, inventories is equal to inventory days on hand $\times$ (COGS/365), and accounts payable is equal to days payable outstanding $\times$ (COGS/365).

Using the data in Exhibit 8 and the revenue and COGS forecasts, year-end working capital account balances are as follows:

Year (CNY millions)	1	2	3	4	5	6
Revenue	174,915	205,839	245,866	290,122	336,541	380,292
COGS	152,723	177,285	209,114	240,801	279,329	319,445
Accounts receivable	5,598	6,949	10,161	11,990	13,908	15,716
Inventories	29,481	32,585	41,671	47,985	55,663	63,657
Accounts payable	46,287	59,528	72,199	83,139	96,442	110,292
Days sales outstanding	12	12	15	15	15	15
Inventory days on hand	70	67	73	73	73	73
Days payable outstanding	111	123	126	126	126	126

While a historical results approach is common for working capital efficiency ratios, analysts can also use the other forecast approaches. Exhibit 9 discusses forecast objects and approaches that Elaine Nguyen can use for Warehouse Club Inc.'s operating costs and working capital.

Exhibit 9: Forecast Approaches for Warehouse Club Inc.'s Operating Costs and Working Capital

Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Gross margin (on net sales)	Historical results	From 2X10 to 2X19, Warehouse's gross margin increased by an average of 2 bps per year. Such an increase may reflect increasing bargaining power over suppliers with scale, and Nguyen can use the same 2 bps per year increase in gross margin as a forecast.
	Historical base rates and convergence	Both the industry average gross margin and that of larger, more mature retailers such as Walmart are much greater than Warehouse's gross margin. But Warehouse's business model is to use a low gross margin on net sales to drive high-margin membership revenues. Therefore, Nguyen probably will not use this approach.
	Management guidance	If Warehouse management provides guidance on margins, Nguyen can use this approach as the forecast, after evaluating management's forecasting credibility.
	Analyst's discretionary forecast	Nguyen may formulate a discretionary forecast for gross margin by exploring the relationship between it, inflation/deflation in Warehouse's product categories, and management's desire for member savings vs. competitors to equal 5× the annual price of membership.
Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Selling, general, and administrative expenses (SG&A) as a percentage of sales	Historical results	SG&A, which captures the cost of operating stores and some merchandise distribution, is primarily composed of employee payroll. Relative to sales, it has remained mostly stable at or around 9%. Nguyen may forecast it to remain the same.
	Historical base rates and convergence	Warehouse's SG&A as a percentage of sales is already lower than that of larger retailer competitors (and substantially lower than the retail industry average), owing to its business model differences that emphasize store productivity and cost leadership. Nguyen will therefore probably not use this approach to forecast SG&A.
	Management guidance	If Warehouse management provides guidance on SG&A, Nguyen can use this approach as the forecast, after evaluating management's forecasting credibility.
	Analyst's discretionary forecast	Since SG&A is primarily composed of employee payroll, Nguyen may forecast it based on projected employee productivity (e.g., employees to stores) and projected wage rates, which may be based on macroeconomic variables like projected economy-wide wage growth.

Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Working capital ratios (e.g., DSO, DOH, DPO)	Historical results	As of 2X19, Warehouse's DSO, DOH, and DPO were 3, 30, and 28, respectively. DSO and DPO have been stable for the last 10 years, while DOH has decreased by 7 days since 2X10. Nguyen might forecast a stable DSO and DPO and a modestly declining DOH over the next several years.
	Historical base rates and convergence	Warehouse's working capital ratios and cash conversion cycle are materially better than the industry average and some larger peers and have been for many years, owing to its business model differences. Nguyen will therefore probably not use this forecast approach, unless Warehouse changes its business model (e.g., it launches an e-commerce business).
	Management guidance	If Warehouse management provides guidance on working capital (e.g., a projection for the net working capital to sales ratio), Nguyen can use this approach as the forecast, after evaluating management's forecasting credibility.
	Analyst's discretionary forecast	Nguyen may develop a discretionary forecast for working capital by, for example, estimating net working capital to sales ratios by product category (e.g., fresh foods, packaged foods, non-food merchandise) and applying them to a net sales forecast also broken down by category.

QUESTION SET



Iliso Marketplace Ltd., the e-commerce retailer introduced in prior modules, reports its GMV by product category: in 2X19, the mix was 55% food and grocery, 20% media and electronics, 8% home goods and décor, and 17% apparel and beauty products. However, Iliso presents only aggregated financial results (no segment disclosures).

Iliso's cost of revenue is primarily composed of the inventory costs of products sold to customers where Iliso records revenue gross. Inventory costs include shipping and handling costs. Additionally, cost of revenue includes outbound shipping and logistics-related expenses, as well as delivery service costs from the third-party merchant business, where Iliso is the delivery service provider. Cost of revenue also includes depreciation expense related to inventory handling and shipping assets.

Iliso's operating, general, and administrative expenses include costs incurred in operating and staffing its fulfillment centers, customer service costs, payment processing fees, costs related to the design, execution, and maintenance of Iliso's technology infrastructure, and general corporate function costs.

Iliso has minimal accounts receivable, because customers pay with credit cards that settle in less than a week. Iliso does have significant inventory, accounts payable, and accrued expenses on its balance sheet.

1. How might an analyst use Iliso's disclosure of GMV by product category?

Solution:

An analyst could estimate cost of sales/sales and gross profit margin by category by looking at other retailers that primarily sell products in the specific category. Information on those companies could be used to check whether Iliso's gross profit margin is consistent with that implied by the product category mix. Also, if the mix is expected to change, an analyst can adjust the aggregate cost of sales/sales and gross profit margins accordingly. The infor-

mation should probably not be used directly to develop detailed forecasts by category, because those forecasts would not be verifiable and would not be an efficient use of the analyst's time.

2. It is *most* appropriate to forecast Iliso's cost of revenue as a percentage of its:

- A. retail sales.
- B. total revenues.
- C. total GMV sales.

Solution:

A is correct. Cost of revenue should be based on Iliso's retail sales. Cost of revenue is unaffected by third-party merchandise sales unless the merchant uses Iliso's fulfillment network for warehousing and delivery. The company does not provide information on such use by third parties, and it would be a small cost compared with cost of revenue for Iliso's retail sales. Iliso's cost of revenue is primarily composed of the inventory costs of products sold to customers where Iliso records revenue gross.

3. Discuss the use of a historical base rates and convergence approach, calculated using other retailers' financial results, for forecasting Iliso's operating expenses.

Solution:

Iliso's e-commerce business model and the breadth of the retail sales industry may make industry averages inappropriate to use for Iliso. However, if there are larger, more mature e-commerce retailers to calculate statistics for, this approach may be appropriate.

4. If food and grocery increases as a proportion of Iliso's retail sales, which of the following would *most likely* increase?

- A. Inventory turnover
- B. Gross profit margin
- C. Current assets/sales

Solution:

A is correct. Inventory turnover would increase because of the perishable nature of food and grocery. Inventory as a percentage of sales would decrease, resulting in a decrease in current assets to sales.

FORECASTING CAPITAL INVESTMENTS AND CAPITAL STRUCTURE

5



explain approaches to forecasting a company's capital investments and capital structure

Projections for long-term assets are based on cash flow statement and income statement projections, because net PP&E and intangible assets on the balance sheet primarily increase due to capital expenditures and decrease due to depreciation and

amortization expenses. Capital expenditures can be broken down into maintenance capital expenditures necessary to sustain the current business and growth capital expenditures needed to expand the business. Maintenance capital expenditure forecasts are often based on historical depreciation and amortization expenses, usually with a small adjustment upward to account for inflation in capital goods. For businesses with low fixed asset turnover, maintenance capital expenditure requirements can be quite high. Growth capital expenditure forecasts are more discretionary and are tied to management's expansion plans and revenue growth. Depreciation and amortization forecasts are based on net PP&E and intangibles on the balance sheet (which increase due to capital expenditures) and their useful lives as assumed by management's accounting policies, which can be approximated by the ratio of gross fixed assets to depreciation and amortization expenses. Information may also be found in the notes to the financial statements.

Analysts must also make projections about a company's future capital structure. Leverage ratios—such as debt to capital, debt to equity, and debt to EBITDA—are often used as the forecast object to project future debt and equity levels. Analysts should consider historical company practice, management's financial strategy, and the capital requirements implied by the capital expenditure assumptions when projecting the future capital structure. Management may provide guidance on target capital structure, debt covenant ratios (e.g., net debt to EBITDA), and capital expenditures, sometimes broken down into maintenance, growth, and acquisitions. Example 3 demonstrates the use of these forecast objects for YY Ltd., the same company in Example 2.

EXAMPLE 3

Balance Sheet Modeling

Exhibit 10 shows financial data for YY Ltd. related to its PP&E and intangible assets. Based on the data in Exhibit 10 and the data and analysis in Example 2, answer Questions 1 and 2.

Exhibit 10: YY Ltd. Long-Term Asset Data (CNY millions)

Year	1	2	3
PP&E, net	5,068	6,992	6,306
Goodwill	282	248	253
Intangible assets, net	1,779	1,424	4,013
Total fixed assets	7,129	8,664	10,572
Capital expenditures – PP&E	3,785	3,405	3,026
Capital expenditures – intangibles	333	142	3,310
Depreciation expense	220	324	518
Amortization expense	529	486	666

Note: PP&E and intangibles asset account balances were also affected each year by changes in exchange rates and by disposals, which are not shown in the exhibit. Assume that such effects are zero in years 4–6.

1. Calculate the following:

- A. Capital expenditures (for both PP&E and intangibles) as a percentage of revenue (for years 1–3)

- B.** Depreciation expense as a percentage of beginning-of-year PP&E, net (for years 2 and 3)
- C.** Amortization expense as a percentage of beginning-of-year intangible assets, net (for years 2 and 3).

Solution:

Using the data, the following percentages were calculated.

Year	1	2	3
Revenue	174,915	205,839	245,866
Capital expenditures – PP&E (% of revenue)	2.2%	1.7%	1.2%
Capital expenditures – intangibles (% of revenue)	0.2%	0.1%	1.3%
Depreciation (% of beginning PP&E)		6%	7%
Amortization (% of beginning intangibles)		27%	47%

2. Given the following assumptions and forecast revenue from Example 2, calculate expected total fixed assets for years 4–6.

- Capital expenditures for PP&E as a percentage of revenue to remain at the year 3 level
- Capital expenditures for intangibles as a percentage of revenue to remain at the year 1 level
- Goodwill to remain at the year 3 level
- Depreciation and amortization expenses as a percentage of beginning-of-year PP&E, net, and intangible assets, net, to remain at year 3 levels

Solution:

Using the data provided, expected total fixed assets for years 4–6 was calculated as CNY12,351 million; CNY15,179 million; and CNY18,662 million, respectively. PP&E, net, for each year was calculated as the prior period balance plus capital expenditures minus depreciation expense. Intangible assets, net, for each year was calculated in a similar fashion. Goodwill was held constant at CNY253 million.

Year	1	2	3	4	5	6
Revenue	174,915	205,839	245,866	290,122	336,541	380,292
PP&E, net	5,068	6,992	6,306	9,410	12,854	16,583
Goodwill	282	248	253	253	253	253
Other intangible assets, net	1,779	1,424	4,013	2,688	2,072	1,827
Total fixed assets	7,129	8,664	10,572	12,351	15,179	18,662
Capital expenditures – PP&E	3,785	3,405	3,026	3,571	4,142	4,680

Year	1	2	3	4	5	6
Capital expenditures – PP&E (% of revenue)	2.2%	1.7%	1.2%	1.2%	1.2%	1.2%
Capital expenditures – intangibles	333	142	3,310	552	641	724
Capital expenditures – intangibles (% of revenue)	0.2%	0.1%	1.3%	0.2%	0.2%	0.2%
Depreciation expense	220	324	518	467	697	952
Depreciation (% of beginning PP&E)		6%	7%	7%	7%	7%
Amortization expense	529	486	666	1,877	1,257	969
Amortization (% of beginning intangibles)		27%	47%	47%	47%	47%

3. Based on Exhibit 11, which shows financial data for YY Ltd. related to its capital structure and profitability, answer Question 3.

Exhibit 11: YY Ltd. Debt and Profitability Data (CNY millions)

Year	1	2	3
Gross debt	10,931	17,624	17,597
Revenue	174,915	205,839	245,866
EBITDA	9,304	12,343	14,190
EBITDA margin	5.3%	6.0%	5.8%

YY Ltd. management has a year 6 gross debt to EBITDA ratio target of 2.0.

- A. Assuming an EBITDA margin of 6.0%, revenue forecasts from Example 2, and gross debt to EBITDA ratios of 1.25, 1.50, and 2.0 for years 4, 5, and 6, respectively, calculate expected gross debt for years 4–6.
- B. Given the results of part A, how much incremental borrowing does the forecast imply from year 3 to year 6?

Solution:

Gross debt of CNY21,579 million; CNY30,289 million; and CNY45,635 million are estimated for years 4–6, respectively, for YY Ltd. This forecast is made by first multiplying forecast revenue by the forecast EBITDA margin to calculate forecast EBITDA. Then, the expected gross debt to EBITDA ratio is multiplied by the forecast EBITDA to calculate forecast gross debt.

Year	1	2	3	4	5	6
Gross debt	10,931	17,624	17,597	21,759	30,289	45,635
Revenue	174,915	205,839	245,866	290,122	336,541	380,292

Year	1	2	3	4	5	6
EBITDA	9,304	12,343	14,190	17,407	20,192	22,818
EBITDA margin	5.3%	6.0%	5.8%	6.0%	6.0%	6.0%
Gross debt to EBITDA	1.17	1.43	1.24	1.25	1.50	2.00

Exhibit 12 discusses forecast objects and approaches that Elaine Nguyen can use for Warehouse Club Inc.'s capital expenditures and gross debt to EBITDA.

Exhibit 12: Forecast Approaches for Warehouse Club Inc.'s Capital Expenditures and Structure

Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Capital expenditures as a percentage of net sales	Historical results	Warehouse has spent 2% of net sales on capital expenditures for the last 10 years, which Nguyen can use as the forecast going forward.
	Historical base rates and convergence	More mature retailers (e.g., Walmart in the USA) have all but stopped opening new stores, which has reduced their capital requirements to maintenance and new initiatives like e-commerce. If Warehouse does not intend to go to new geographies or launch an e-commerce business, Nguyen might forecast capital expenditures trending down to maintenance level, along with a decline in new-store openings per year, as Warehouse matures.
	Management guidance	Warehouse management provides guidance on new-store openings, which Nguyen can use to forecast capital expenditures (along with a maintenance capital expenditure forecast and a forecast for growth capital expenditures per new store).
	Analyst's discretionary forecast	Nguyen might instead use a discretionary forecast for capital expenditures by projecting a maintenance CAPEX amount (not disclosed by management) and growth CAPEX associated with new-store openings.

Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Gross debt to EBITDA	Historical results	For the last five years, Warehouse's ratio of debt to trailing twelve months EBITDA has been 1.10, which Nguyen can project going forward. The product of this ratio and projected EBITDA (from earlier forecasts of sales and operating expenses) is Warehouse's gross debt.
	Historical base rates and convergence	Warehouse's more mature competitors and peers use a greater amount of debt in their capital structure, and return more capital to shareholders with buybacks and dividends. Nguyen might forecast the same for Warehouse over time, with an increasing ratio of debt to EBITDA and percentage of free cash flow returned to shareholders.
	Management guidance	Warehouse's management may provide a target capital structure through a target debt to EBITDA ratio or credit rating, which Nguyen can use as the forecast.
	Analyst's discretionary forecast	Nguyen might instead make a discretionary forecast linked to her free cash flow forecast, projecting a debt to EBITDA ratio and return of capital to shareholders. Depending on Fyleton Investments' strategy, Nguyen might engage with Warehouse's management to advocate for such a capital structure.

QUESTION SET

An analyst gathers the following information related to Iliso's capital expenditures and capital structure (USD millions):



	2X15	2X16	2X17	2X18	2X19
Iliso's retail sales	2,589	3,424	5,100	6,052	7,336
Operating income	−1	18	61	72	130
Depreciation expense	25	43	65	91	123
PP&E (net)	198	294	435	590	768
Short-term debt	56	255	198	66	0
Long-term debt	43	157	168	144	144
Equity	548	732	858	1,006	1,178
Capital expenditures	79	140	208	247	301

1. Which of the following would the analyst *most likely* use in estimating Iliso's maintenance capital expenditures?

A. Operating income
 B. Depreciation expense
 C. Earnings before interest, taxes, depreciation, and amortization (EBITDA)

Solution:
 B is correct. Depreciation expense can serve as the basis for maintenance capital expenditures, as it is management's estimate of the cost of fixed assets expensed on the income statement in a manner that tracks its use.

2. Which of the following approaches is *most* appropriate for forecasting Iliso's capital expenditures?

A. Historical results
 B. Analyst's discretionary forecast
 C. Historical base rates and convergence

Solution:
 A is correct. Capital expenditures have been 4.1% of retail sales for the past four years (2X16–2X19). The historical results approach—using the bottom-up driver retail sales—is most appropriate. An analyst's discretionary forecast is not required, because Iliso is not in a cyclical industry or undergoing a fundamental change. Iliso's unique model and the breadth of the industry make the approach using historical base rates and convergence to the industry inappropriate.

3. The analyst forecasts 2X20 total revenues of 8,411 million, operating income of 2% of total revenues, capital expenditures of 4% of total revenues, depreciation expense of 20% of opening PP&E (net), and debt/EBITDA of 0.50. The analyst's forecast for debt (in millions) is *closest* to:

A. 84.
 B. 161.

C. 195.

Solution:

B is correct.

$$2X20 \text{ operating income} = 0.02(8,411) = 168$$

$$2X20 \text{ depreciation expense} = 0.20(768) = 154$$

$$\text{EBITDA} = 168 + 154 = 322$$

$$\text{Debt} = 0.50(322) = 161$$

4. The analyst notes that Iliso's common stock and paid-in capital accounts increased, in total, by 75 million from 2X18 to 2X19, but only 42 million was reported as stock issuance on the statement of cash flows. Retained earnings increased by the amount of net income. Which of the following *most likely* explains the difference of 33 million?

- A. A stock split
- B. A stock dividend
- C. Stock-based compensation

Solution:

C is correct. Stock-based compensation is a non-cash expense in the income statement and is reported as an addition to net income in the indirect method of reporting cash flow from operations. The issuance also increases common stock and paid-in capital. A stock split typically simply increases the number of shares outstanding and decreases any stated par value per share. A stock dividend increases common stock and paid-in capital and reduces retained earnings by the same amount.

6

SCENARIO ANALYSIS



describe the use of scenario analysis in forecasting

Recall from earlier lessons that industry and business risks may result in future outcomes that differ from expectations. The final step in forecasting involves incorporating the possibility of different outcomes based on key risk factors as well as judging their likelihood of occurrence. Generic risk factors that affect all companies—but to varying degrees—include changes in the business cycle, competition, inflation and deflation, and technological developments. Rather than develop single point estimate forecasts, analysts make several forecast scenarios that vary based on different outcomes with respect to key risk factors. Investors compare these scenarios with other analysts' (e.g., sell-side analysts) forecasts for a company, as well as forecasts implied by current valuations, to make investment decisions.

We demonstrate the use of scenario analysis for technological developments that threaten to cannibalize demand for an existing product in the following historical case study. Technological developments can affect demand for a product, the quantity supplied of a product, or both. When changes in technology lead to lower manufacturing

costs, the supply curve will shift to the right as suppliers produce more of the product at the same price. Conversely, if changes in technology result in the development of attractive substitute products, the demand curve will shift to the left.

EXAMPLE 4

Quantifying the Tablet Market's Potential to Cannibalize Demand for Personal Computers

The worldwide tablet market experienced a major technological development with the introduction of the Apple iPad tablet in April 2010, which was expected to have (and indeed did have) important implications for the manufacturers of desktop and laptop computers. A tablet promised to offer the capabilities of a portable personal computer (PC) with a touchscreen interface instead of a keyboard. Another distinguishing feature of the then-new tablets was that—unlike the majority of PCs, which run on the Microsoft Windows platform—they would run on a non-Microsoft operating system: Apple's iOS or Google's Android. Given the tablet's ability to perform many of the most common tasks of a PC—including emailing, browsing the web, sharing photos, playing music, watching movies, playing games, keeping a calendar, and managing contacts—an analyst at that time might reasonably have wondered to what extent sales of tablets might cannibalize demand for PCs and what the potential impact might be on Microsoft's sales and earnings. Exhibit 13 presents one approach to answering these questions. It is set at the start of 2012, just over a year after the launch of the iPad. It is presented from the position of an analyst assessing the impact of the tablet on the PC market and Microsoft.

Exhibit 13: Unit and Revenue Projections (in thousands, unless noted)

Pre-Cannibalization PC Projections	FY2011	FY2012E	FY2013E	FY2014E	3-Year CAGR
Consumer PC shipments	170,022	174,430	184,120	193,811	4.5%
Non-consumer PC shipments	180,881	185,570	195,880	206,189	4.5%
Total global PC shipments	350,903	360,000	380,000	400,000	4.5%
% of which is consumer	48%	48%	48%	48%	
% of which is non-consumer	52%	52%	52%	52%	
Consumer tablet shipments	36,785	82,800	111,250	148,750	59.3%
Non-consumer tablet shipments	1,686	7,200	13,750	26,250	149.7%
Total global tablet shipments	38,471	90,000	125,000	175,000	65.7%
% of which is consumer	96%	92%	89%	85%	
% of which is non-consumer	4%	8%	11%	15%	

Pre-Cannibalization PC Projections	FY2011	FY2012E	FY2013E	FY2014E	3-Year CAGR
Cannibalization factor, consumer	30%	30%	30%	30%	
Cannibalization factor, non-consumer	10%	10%	10%	10%	
# of consumer PCs cannibalized by tablets	11,036	24,840	33,375	44,625	
# of non-consumer PCs cannibalized by tablets	169	720	1,375	2,625	
Total PCs cannibalized by tablets	11,205	25,560	34,750	47,250	
% of total PCs cannibalized by tablets	3.2%	7.1%	9.1%	11.8%	
Post-Cannibalization PC Projections					
Consumer PC shipments	158,987	149,590	150,745	149,186	-2.1%
Non-consumer PC shipments	180,712	184,850	194,505	203,564	4.0%
Total global PC shipments	339,699	334,440	345,250	352,750	1.3%
<i>Microsoft implied average selling price (USD)</i>					
Consumer	85	85	85	85	
Non-consumer	155	155	155	155	
<i>Revenue impact for Microsoft (USD millions)</i>					
Consumer	938	2,111	2,837	3,793	
Non-consumer	26	112	213	407	
Total revenue impact	964	2,223	3,050	4,200	

Note: CAGR is compound annual growth rate. Non-consumer includes enterprise, education, and government purchasers.

Sources: Based on data from Gartner, JPMorgan, Microsoft, and authors' analysis.

Worldwide market shipments of PCs in FY2011 totaled 350.9 million units, while worldwide shipments of tablets totaled 38.5 million units (source: Gartner Personal Computer Quarterly Statistics Worldwide Database). Shipments to consumers represented 96% of total tablet shipments during fiscal year 2011. We next estimate the magnitude of the potential substitution effect, or cannibalization factor, that tablets will have on the PC market. Because the cannibalization factor depends on many different variables—including user preferences, end-use application, and whether the purchaser already owns a PC—we use a range of potential estimates. Moreover, we also divide the worldwide PC market into consumer and non-consumer (enterprise, education, and government purchasers), because the degree of substitution is likely to differ between the two. For purposes of illustration, we assume a cannibalization factor of 30% for the consumer market and 10% for the non-consumer market in our base case scenario.

In addition, the base case scenario assumes that non-consumer adoption of tablets increases to 15% of the market from 4% in 2011. Moreover, although the composition of the global PC market is roughly evenly divided between consumers and non-consumers (48% and 52% in fiscal year 2011, respectively), the non-consumer segment is significantly more profitable for Microsoft, because approximately 80% of the company's Office products are sold to enterprise, education, and government institutions. The average selling price (ASP) estimates are derived by dividing Microsoft's estimated average revenue by customer type for the prior three years by Microsoft's estimated PC shipments for each type of customer. By multiplying the projected number of PCs cannibalized by tablets by the estimated ASP, we can derive an estimate of the revenue impact for Microsoft. For example, in FY2012, it is projected that 24.8 million consumer PCs will be cannibalized by sales of tablets. With a consumer ASP of USD85, this cannibalization implies a revenue loss for Microsoft of USD2.1 billion (24.8 million units $\times$ USD85 per unit = USD2.1 billion).

Once the revenue impact has been projected, the next step is to estimate the impact of lower PC unit volumes on operating costs and margins. We begin by analyzing the cost structure of Microsoft and, more specifically, the breakdown between fixed and variable costs. Most software companies have a cost structure with a relatively high proportion of fixed costs and a low proportion of variable costs, because costs related to product development and marketing (mostly fixed) are sunk and unrecoverable, whereas the cost of producing an additional copy of the software (mostly variable) is relatively low. Because very few, if any, companies provide an explicit breakdown of fixed versus variable costs, an estimate almost always needs to be made. One method is to use the formula

$$\% \Delta (\text{Cost of revenue} + \text{Operating expenses}) / \% \Delta \text{ revenue},$$

where $\% \Delta$ is "percentage change in," used as a proxy for variable cost percentage. Another approach is to assign an estimate of the percentage of fixed and variable costs to the various components of operating expenses. The two approaches are illustrated in Exhibit 14 and Exhibit 15.

**Exhibit 14: Estimation of Variable Costs for Microsoft, Method 1
(USD millions)**

Selected Operating Segments	FY2009	FY2010	FY2011	FY2011/FY2009 Percentage Change
Revenue:				
Windows and Windows Live	15,563	18,792	18,778	
Microsoft business division	19,211	19,345	21,986	
Total segment revenue	34,774	38,137	40,764	17%
Operating expenses:				
Windows and Windows Live	6,191	6,539	6,810	

Selected Operating Segments	FY2009	FY2010	FY2011	FY2011/FY2009 Percentage Change
Microsoft business division	8,058	7,703	8,159	
Total operating expenses	14,249	14,242	14,969	5%

%Variable cost estimate $\approx$ % Δ (Cost of revenue + Operating expenses)/% Δ revenue $\approx$ 5%/17% $\approx$ **29%**

%Fixed cost $\approx$ 1 – %Variable cost $\approx$ 1 – 29% $\approx$ **71%**

Exhibit 15: Estimation of Variable Costs for Microsoft, Method 2 (USD millions)

Operating Expenses	FY2009	FY2010	FY2011	FY2009–FY2011 Average	% of Total Op. Expenses	Estimated % of Fixed Costs	Fixed Cost Contribution
Cost of revenue (excl. depreciation)	10,455	10,595	13,577	11,542	29%	20%	6%
Depreciation expense	1,700	1,800	2,000	1,833	5%	100%	5%
Total cost of revenue	12,155	12,395	15,577	13,376	34%		10%
R&D	9,010	8,714	9,043	8,922	22%	100%	22%
Sales and marketing	12,879	13,214	13,940	13,344	34%	80%	27%
General and admin.	4,030	4,063	4,222	4,105	10%	100%	10%
Total operating expenses	38,074	38,386	42,782	39,747	100%		60%
Estimated percentage of Microsoft's total cost structure that is fixed:							70%

Note: Fiscal year ends in June.

Sources: Microsoft 2011 Form 10-K and authors' analysis.

As can be seen, Microsoft's cost structure appears to consist of approximately 70% fixed costs and 30% variable costs. Note, however, that a growing company like Microsoft will typically re-invest in PP&E to support future growth, so even those expenses that appear to be "fixed" will increase over time. To adjust for this expected growth in fixed costs, this example includes an assumption that the change in fixed costs will be half the rate of the change in sales. Variable costs are projected to change at the same rate as sales. As shown in Exhibit 16, after incorporating these assumptions into the projections, an assumed 7.0% compound annual growth rate (CAGR) in revenue through FY2014 would translate into a 10.6% CAGR in operating income $[(36,757/27,161)^{1/3} - 1 = 0.106$, or 10.6%]. In addition, these assumptions would result in an operating margin expansion of 410 bps over the same period $(42.9\% - 38.8\% = 4.1$ pps, or 410 bps), because of the significant amount of operating leverage that exists as a result of

a relatively large fixed cost base. With the further assumptions of no change in other income, a constant effective tax rate, and no change in shares outstanding, the pre-cannibalization model, shown in Exhibit 16, results in projected revenue of USD85.7 billion, operating income of USD36.8 billion, an operating margin of 42.9%, and EPS that increases at a CAGR of 10.3% to USD3.62 in FY2014.

**Exhibit 16: Microsoft Pre-Cannibalization EPS Projections
(USD millions, except EPS)**

	FY2011	FY2012E	FY2013E	FY2014E	3-Year CAGR
Revenue	69,943	74,839	80,078	85,683	7.0%
YoY%		7.0%	7.0%	7.0%	
<i>Operating Expenses</i>					
Fixed (70%)	29,947	30,996	32,080	33,203	3.5%
Variable (30%)	12,835	13,733	14,694	15,723	7.0%
Total operating expenses	42,782	44,729	46,774	48,926	4.6%
Operating income	27,161	30,110	33,303	36,757	10.6%
Operating margin	38.83%	40.23%	41.59%	42.90%	
Other income (expense)	910	910	910	910	
Pre-tax income	28,071	31,020	34,213	37,667	
Provision for income taxes	4,921	5,438	5,998	6,603	
Effective tax rate	17.53%	17.53%	17.53%	17.53%	
Net income	23,150	25,582	28,215	31,064	
Weighted average shares outstanding, diluted	8,593	8,593	8,593	8,593	
Estimated EPS pre-cannibalization	2.69	2.98	3.28	3.62	10.3%

In the post-cannibalization scenario, as shown in Exhibit 17, revenue is reduced each year to reflect the expected impact of cannibalization, which results in a decrease in the CAGR of revenue over the period to 5.2%, down from 7.0% in the pre-cannibalization scenario. Given the reduction in revenue growth and holding the cost structure constant at 70/30 fixed versus variable costs, operating income growth slows to a CAGR of 8.0%, down from 10.6% in the pre-cannibalization scenario. Operating margin at the end of the period is reduced by approximately 100 bps, from 42.9% to 41.9%, because the company is unable to leverage its fixed cost base to the same degree as a result of slower revenue growth. Overall, in the post-cannibalization scenario, Microsoft is expected to generate revenue of USD81.5 billion, operating income of USD34.2 billion, an operating margin of 41.9%, and EPS that increases at a CAGR of 7.7%

to USD3.37 in FY2014. Thus, the cannibalization of PCs as a result of projected growth in the tablet market is expected to reduce the company's annual revenues in FY2014 by USD4.2 billion, operating income by USD2.6 billion, operating margins by 96 bps, and EPS by USD0.25.

Exhibit 17: Microsoft Post-Cannibalization EPS Projections, Base Case Scenario (USD millions, except EPS and unless noted)

	FY2011	FY2012E	FY2013E	FY2014E	3-Year CAGR
Revenue	69,943	72,616	77,028	81,483	5.2%
YoY%		3.8%	6.1%	5.8%	
<i>Operating Expenses</i>					
Fixed (70%)	29,947	30,520	31,447	32,356	2.6%
Variable (30%)	12,835	13,325	14,135	14,952	5.2%
Total operating expenses	42,782	43,845	45,581	47,308	3.4%
Operating income	27,161	28,771	31,446	34,175	8.0%
Operating margin	38.83%	39.62%	40.82%	41.94%	
Other income (expense)	910	910	910	910	
Pre-tax income	28,071	29,681	32,356	35,085	
Provision for income taxes	4,921	5,203	5,672	6,151	
Effective tax rate	17.53%	17.53%	17.53%	17.53%	
Net income	23,150	24,478	26,684	28,934	
Weighted average shares outstanding, diluted	8,593	8,593	8,593	8,593	
Estimated EPS post-cannibalization	2.69	2.85	3.11	3.37	7.7%
Compared with pre-cannibalization:					
Estimated impact on operating margin		-61 bps	-77 bps	-96 bps	
Estimated impact on EPS (USD)		-0.13	-0.17	-0.25	-2.6%

KNOWLEDGE CHECK



Answer the following questions using the data from the exhibits on Microsoft:

1. Estimate post-cannibalization global PC shipments in FY2012 assuming a cannibalization factor of 40% for consumers and 15% for non-consumers.

Solution:

The number of PCs cannibalized by tablets is equal to the product of the expected number of global tablet shipments, the percentage representation of each category, and the cannibalization factor for the category. Tablet

shipments in FY2012 are projected to be 90 million units; (90 million tablets $\times$ 92% consumer representation $\times$ 40% consumer cannibalization factor = 33.12 million consumer PCs cannibalized by tablets) + (90 million tablets $\times$ 8% non-consumer representation $\times$ 15% cannibalization = 1.08 million non-consumer PCs cannibalized by tablets) = 34.2 million total PCs cannibalized by tablets. Post-cannibalization shipments are equal to pre-cannibalization shipments minus expected cannibalization, or 360 million – 34.2 million = 325.8 million.

2. Using the results derived in Question 1, estimate the post-cannibalization revenue in FY2012 for Microsoft.

Solution:

The estimated impact on revenue is equal to the product of the number of PCs cannibalized and the ASP. Using the results obtained in Question 1 and the ASP data, the expected revenue impact can be calculated as (33.12 million consumer PCs cannibalized by tablets $\times$ USD85 ASP = USD2.815 billion) + (1.08 million non-consumer PCs cannibalized by tablets $\times$ USD155 ASP = USD167.4 million) = USD2.983 billion total impact on revenue for Microsoft. Post-cannibalization revenue is equal to pre-cannibalization revenue minus the estimated impact of cannibalization on revenue, or USD74.839 billion – USD2.983 billion = USD71.856 billion.

3. Using the estimate for post-cannibalization revenue derived in Question 2 and the cost structure provided, estimate post-cannibalization operating income and operating margin in FY2012 for Microsoft. Assume that fixed costs change at half the rate of the change in sales.

Solution:

Exhibit 18: Solution to Question 3 (USD millions)

	FY2011	FY2012E	Notes
Revenue	69,943	71,856	Derived from Question 2
YoY%		2.74%	Rate of change in sales used to estimate operating expenses
<i>Operating Expenses</i>			
Fixed (70%)	29,947	30,357	Fixed costs change at half the rate of the change in sales, or $29,947 \times (1 + 2.74\%/2)$
Variable (30%)	12,835	13,186	Variable costs change at the same rate as the change in sales, or $12,835 \times (1 + 2.74\%)$
Total operating expenses	42,782	43,543	Although not shown, operating expenses include COGS

	FY2011	FY2012E	Notes
Operating income	27,161	28,313	Revenue minus total operating expenses, or $71,856 - 43,543 = 28,313$
Operating margin	38.8%	39.4%	Operating income divided by revenue, or $28,313/71,856 = 39.4\%$

Post-cannibalization operating income and operating margin in FY2012 for Microsoft are USD28,314 million and 39.4%, respectively.

4. Using the estimate for operating income derived in Question 3 and the data in the exhibits, calculate the expected post-cannibalization EPS in FY2012 for Microsoft. Assume that other income (expense), the effective tax rate, and the diluted weighted average shares outstanding provided for FY2011 remain constant in FY2012.

Solution:

Exhibit 19: Solution to Question 4 (USD millions, except EPS and unless noted)

	FY2011	FY2012E	Notes
Revenue	69,943	71,856	
YoY%		2.74%	
<i>Operating Expenses</i>			
Fixed (70%)	29,947	30,357	
Variable (30%)	12,835	13,186	
Total operating expenses	42,782	43,543	
Operating income	27,161	28,314	
Operating margin	38.8%	39.4%	
Other income (expense)	910	910	
Pre-tax income	28,071	29,224	Operating income + Other income (expense), or $28,314 + 910 = 29,224$
Provision for income taxes	4,921	5,123	Pre-tax income $\times$ Effective tax rate, or $29,224 \times 17.53\% = 5,123$
Effective tax rate	17.53%	17.53%	
Net income	23,150	24,101	Pre-tax income – Provision for income taxes, or $29,224 - 5,123 = 24,101$

	FY2011	FY2012E	Notes
Weighted average shares outstanding, diluted	8,593	8,593	
Estimated EPS post-cannibalization (USD)	2.69	2.80	Net income/Wtd. avg. shs. out., or 24,101/8,593 = USD2.80

Whenever one is estimating something that depends on many different variables that are difficult to measure, we recommend altering some of the assumptions to generate a range of estimates based on various scenarios. Thus, having developed a forecast under a base case cannibalization scenario, we can analyze the sensitivity of the results by altering the cannibalization assumptions. The base case scenario corresponds to the assumptions reflected in the boxed center of the table in Exhibit 20. Exhibit 21 summarizes the results of both bull and bear case scenarios, showing the estimated FY2014 EPS under alternative estimated cannibalization factors.

Exhibit 20: Estimated FY2014 EPS Sensitivity to Changes in Cannibalization Rates (USD)

		Non-Consumer Cannibalization				
		0%	5%	10%	15%	20%
Consumer Cannibalization	15%	-0.11	-0.12	-0.14	-0.15	-0.16
	20%	-0.15	-0.16	-0.17	-0.19	-0.20
	25%	-0.19	-0.20	-0.21	-0.22	-0.23
	30%	-0.22	-0.24	-0.25	-0.26	-0.27
	35%	-0.26	-0.27	-0.28	-0.30	-0.31
	40%	-0.30	-0.31	-0.32	-0.33	-0.35
	45%	-0.34	-0.35	-0.36	-0.37	-0.38

Exhibit 21: Post-Cannibalization EPS Projections for Bull and Bear Case Scenarios (USD millions, except EPS and unless noted)

Bull Case Scenario (Cannibalization Factor: 15% Consumer/5% Non-Consumer)					
	FY2011	FY2012E	FY2013E	FY2014E	3-Year CAGR
Revenue	69,943	73,728	78,553	83,583	6.1%
YoY%		5.4%	6.5%	6.4%	
<i>Operating Expenses</i>					
Fixed (70%)	29,947	30,758	31,764	32,781	3.1%
Variable (30%)	12,835	13,529	14,414	15,338	6.1%
Total operating expenses	42,782	44,287	46,178	48,119	4.0%

Bull Case Scenario (Cannibalization Factor: 15% Consumer/5% Non-Consumer)					
	FY2011	FY2012E	FY2013E	FY2014E	3-Year CAGR
Operating income	27,161	29,441	32,374	35,464	9.3%
Operating margin	38.83%	39.93%	41.21%	42.43%	
Other income (expense)	910	910	910	910	
Pre-tax income	28,071	30,351	33,284	36,374	
Provision for income taxes	4,921	5,321	5,835	6,377	
Effective tax rate	17.53%	17.53%	17.53%	17.53%	
Net income	23,150	25,030	27,449	29,998	
Weighted average shares outstanding, diluted	8,593	8,593	8,593	8,593	
Estimated EPS post-cannibalization (USD)	2.69	2.91	3.19	3.49	9.0%
Compared with pre-cannibalization:					
Estimated impact on operating margin		-30 bps	-38 bps	-47 bps	
Estimated impact on EPS (USD)		-0.07	-0.09	-0.13	-1.3%
Bear Case Scenario (cannibalization factor: 40% consumer, 20% non-consumer)					
	FY2011	FY2012E	FY2013E	FY2014E	3-Year CAGR
Revenue	69,943	71,801	75,869	79,812	4.5%
YoY%		2.7%	5.7%	5.2%	
<i>Operating Expenses</i>					
Fixed (70%)	29,947	30,345	31,205	32,016	2.3%
Variable (30%)	12,835	13,175	13,922	14,646	4.5%
Total operating expenses	42,782	43,520	45,127	46,662	2.9%
Operating income	27,161	28,280	30,742	33,151	6.9%
Operating margin	38.83%	39.39%	40.52%	41.54%	
Other income (expense)	910	910	910	910	
Pre-tax income	28,071	29,190	31,652	34,061	
Provision for income taxes	4,921	5,117	5,549	5,971	
Effective tax rate	17.53%	17.53%	17.53%	17.53%	
Net income	23,150	24,073	26,103	28,090	

Bear Case Scenario (cannibalization factor: 40% consumer, 20% non-consumer)

	FY2011	FY2012E	FY2013E	FY2014E	3-Year CAGR
Weighted average shares outstanding, diluted	8,593	8,593	8,593	8,593	
Estimated EPS post-cannibalization (USD)	2.69	2.80	3.04	3.27	6.7%
Compared with pre-cannibalization:					
Estimated impact on operating margin		-84 bps	-107 bps	-136 bps	
Estimated impact on EPS (USD)		-0.18	-0.24	-0.35	-3.6%

QUESTION SET

1. For which of the following companies is it *most* crucial for an analyst to conduct scenario analysis to estimate a decline in earnings from a recession?

- A. A bank
- B. A chain of large grocery stores
- C. A provider of residential electricity

Solution:

A is correct, because both lending volume and interest rates typically fall in recessions, which results in declining earnings for banks. Demand for both food-at-home and residential electricity is generally not as sensitive to changes in the business cycle.

2. An analyst is preparing scenarios for a company's gross margin. Selected financial data from the prior period include the following:

Average selling price per product EUR6.50

Quantity sold (000s) 2,300

Total revenues (EUR 000s) 14,950

Cost of sales (EUR 000s) 7,475

Gross profit (EUR 000s) 7,475

Gross profit margin 50%

If input prices increase by 20%, the analyst expects the company to increase selling prices by 10%, resulting in a 5% decrease in volume sold. The forecast decrease in gross profit margin is *closest* to:

- A. 4.5%
- B. 7.4%.

C. 15.0%

Solution:

A is correct.

$$\text{Average selling price} = \text{EUR}6.50(1.10) = \text{EUR}7.15$$

$$\text{Quantity sold in 000s} = 2,300(0.95) = 2,185$$

$$\text{Average input cost} = (7,475/2,300)(1.20) = \text{EUR}3.90$$

$$\text{Total revenues in 000s} = 2,185(\text{EUR}7.15) = 15,622.75$$

$$\text{Gross profit in 000s} = 2,185(\text{EUR}7.15 - \text{EUR}3.90) = 7,101.25$$

Gross profit margin

$$= 7,101.25/15,622.75 \approx 45.5\%, \text{ which is } 4.5\% \text{ less than } 50\%$$

3. The invention of a new technology that leads to lower product manufacturing costs in a highly competitive industry is *most likely* to result in:

- A.** lower gross profit margins.
- B.** higher gross profit margins.
- C.** no impact on gross profit margins.

Solution:

C is correct. In a highly competitive industry, savings in manufacturing costs are likely to be passed on to customers through competition in the form of lower prices.

REFERENCES

- Beyer, Anne, Daniel A. Cohen, Thomas Z. Lys, Beverly R. Walther. 2010. "The financial reporting environment: Review of the recent literature." December. *Journal of Accounting and Economics*, Elsevier 50 (2-3): 296–343. 10.1016/j.jacceco.2010.10.003
- Ciconte, Will, Marcus Kirk, Jenny Wu Tucker. 2013. Does the Midpoint of Range Earnings Forecasts Represent Managers' Expectations? (February 1, 2013). *Review of Accounting Studies*, 19(2): 628-660., Available at SSRN: <https://ssrn.com/abstract=2061440> or
- Hutton, A. P., L. F. Lee, S. Z. Shu. 2012. "Do Managers Always Know Better? The Relative Accuracy of Management and Analyst Forecasts." *Journal of Accounting Research* 50:1217–44. 10.1111/j.1475-679X.2012.00461.x

PRACTICE PROBLEMS

1. Forecast objects should be:
 - A. based on independent third-party reports.
 - B. based on individual discrete items, not aggregations.
 - C. disclosed regularly or based on items that are disclosed regularly.
2. Management guidance is *most* appropriately used by analysts in forecasting:
 - A. capital expenditures of a company.
 - B. pricing of commodities sold by a company.
 - C. revenue of a company sensitive to the business cycle.
3. Which of the following is an example of a top-down driver of revenues for a passenger airline?
 - A. Market share of routes
 - B. Number of planes in fleet
 - C. Average ticket price per mile flown
4. An analyst forecasts a company's sales using a historical results approach and top-down drivers (expected industry sales and expected market share). The analyst observes that the forecast is within the range of management guidance on sales but is toward the upper end and not closer to the midpoint. Which of the following is the *most likely* explanation for this observation?
 - A. The analyst has been overly optimistic in estimating the company's expected market share.
 - B. Management's true expectations are toward the upper end, but management has created a range using a pessimistic lower bound that is more easily cleared.
 - C. The midpoint represents management's true expectations, and management has a lower and better estimate of market growth based on macroeconomic variables.
5. A clothing company, which initially produced designer clothing, has increasingly entered into the mass market. Which of the following approaches to forecasting revenues is *least* appropriate?
 - A. Historical results
 - B. Analyst's discretionary forecast
 - C. Historical base rates and convergence
6. An analyst is forecasting operating costs for a company with relatively high fixed costs, sensitivity to economic conditions, and commodity inputs with volatile pricing. The company does not follow a hedging strategy for commodity purchases but tries to buy when prices are low. Which of the following is *most* appropri-

ate to use in forecasting operating costs? The analyst uses:

- A. analyst discretion to forecast all operating costs.
 - B. management guidance to forecast all operating costs.
 - C. management guidance to forecast fixed operating costs and analyst discretion to forecast variable operating costs.
7. An analyst notes that a company's capital expenditures do not follow a discernible pattern; the company seems to have periods of very low capital expenditures and periods of high capital expenditures. Management does not provide any guidance on capital expenditures. The analyst should develop a forecast of capital expenditures based on:
- A. the company's usage of PP&E capacity.
 - B. the industry's average capital expenditures.
 - C. the company's average capital expenditures.
8. An analyst predicts that if a company's technological developments are a success, the company's operating costs will be reduced by 15%. As a result of the reduction in costs, the company will reduce the average selling price of its products by 5% and the volume of sales will increase by 8%. The company's current gross profit margin is 40%. If technological developments occur, the company's gross profit margin will be *closest* to:
- A. 44.8%.
 - B. 46.3%.
 - C. 47.5%.

SOLUTIONS

1. C is correct. Forecast objects should either be disclosed regularly or be directly calculable using what is disclosed regularly. Information that is *not* disclosed regularly (such as a third-party consultancy's report) is suitable to *inform* forecasts but can be problematic for direct use, because forecasts cannot be confirmed in a timely manner. While it may be intuitive to forecast individual discrete items, such as sales and gross margin by individual product line, if gross margin is disclosed only on a consolidated basis, it will be difficult for analysts to verify their product-line gross margin estimates.
2. A is correct. Management has an advantage at the firm level and tends to make more accurate forecasts for objects that are subject to management actions, such as capital expenditures. Management does *not* have an informational advantage over investors in forecasting macroeconomic variables, such as GDP, the business cycle, or the prices of commodities. Hutton et al. (2012) found that investors have an informational advantage over management at the macroeconomic level.
3. A is correct. Market share of routes is a top-down driver of revenues and can be used to estimate an airline's market share. The number of planes in fleet and the average ticket price per mile flown are bottom-up drivers of revenues.
4. B is correct. Management has a tendency to pad its guidance to make it easier to achieve the forecast results for which management will be rewarded. Management does not have an informational advantage at the macroeconomic level.
5. A is correct. Historical results is least appropriate, because the company has changed its competitive strategy. Analyst's discretionary forecast may be appropriate, because the company is undergoing a fundamental change. The company is growing and may be transitioning to be more like the industry, in which case historical base rates and convergence may be an appropriate approach.
6. C is correct. Management has an advantage in forecasting for objects that are subject to its actions (such as capital expenditures and inventories, which affect fixed costs). The analyst is likely to have an informational advantage when it comes to forecasting economic conditions and commodity prices, which affect revenues and variable costs.
7. A is correct. Based on the company's spending pattern, it most likely makes capital expenditures based on capacity needs as it grows. If it is approaching full usage of existing capacity, it will expand.
8. B is correct.

$$\text{Sales} = 100(0.95)(1.08) = 102.6$$

$$\text{Cost of sales} = 60(0.85)(1.08) = 55.1$$

$$\text{Gross profit} = 102.2 - 55.1 = 47.5$$

$$\text{Gross profit margin} = 47.5/102.6 = 46.3\%$$